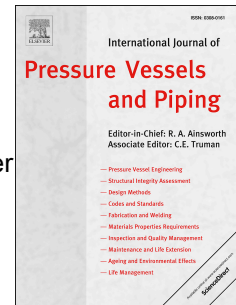


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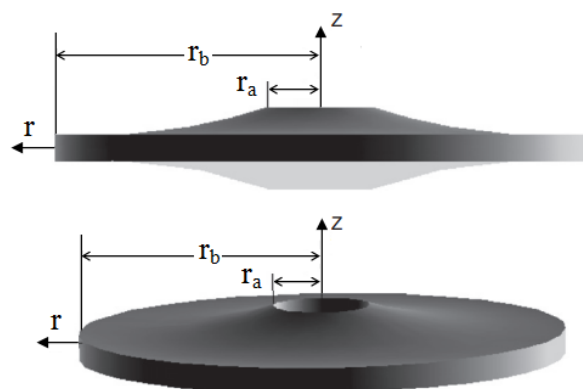
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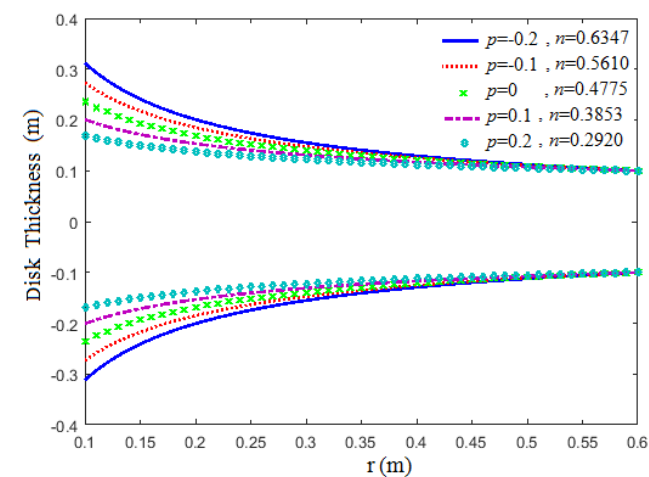
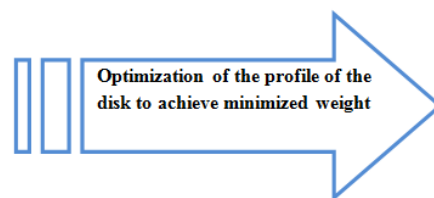
Optimization of weight of functionally graded rotating disks of variable thickness under thermoelastic load

Mohammad Khorsand, Youhong Tang

Graphical abstract



FG rotating disk of varied thickness



Results of the optimized FG rotating disk

Design functionally graded rotating disks under thermoelastic loads: Weight optimization

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Abstract: In this work, an advanced algorithm is employed to optimize the weight of a functionally graded hollow circular disk of varied thickness rotating at a constant angular velocity about its central axis. The disk is under thermoelastic load conditions. The material properties, encompassing elastic modulus thermal expansion coefficient, thermal conductivity, and density and thickness of the disk are assumed to be graded in the radial direction, and the Poisson's ratios are assumed to be constant. A combination of a co-evolutionary particle swarm optimization (*CPSO*) approach coupled with a differential quadrature (*DQ*) method is applied to obtain minimized stress and displacement fields through the geometry of the disk. The role of *DQ* is discretizing and solving the governing equations including motion and boundary conditions existing in the goal function of the *CPSO* method. In comparisons with other optimization results in the literature, the weight reduction results of this algorithm show much superior responses, demonstrating the effectiveness of this algorithm for optimizing constrained complex problems in engineering structures. Also, in numerical simulation, a study is performed to determine the effects of angular velocity and gradient index variations on the distribution of stress and displacement fields.

Keywords: Rotating disk, Thermo-elasticity, Functionally graded materials, Optimization

1. Introduction

Rotating disks are used in many industrial parts encompassing shrink fits, gears, rotors, and flywheels. Thermoelastic analysis of rotating disks is vital because these mechanical parts usually work under high thermal loads and large centrifugal forces. Some attempts have been made to present analytical solutions for disks with constant thickness for elastic and linearly hardening materials [1-4]. On the other hand, functionally graded materials (FGMs) are composites with microstructural details that vary spatially through nonuniform distribution of the reinforcement phase(s). Reinforcements with different properties, sizes, and shapes are used in a continuous manner, resulting in a microstructure with continuously varying thermal and mechanical properties at the macroscopic or continuum scale. Much research has been conducted to describe the behavior of FG structures under thermo-mechanical excitation [5-9]. Hojjati and Jafari [10-11] employed the homotopy perturbation method and Adomian's decomposition method to obtain distributions of stress and displacement fields in rotating annular elastic disks with both uniform and variable thickness and density. By applying variable material properties and using the Runge-Kutta method, Hojjati and Hassani [12] studied the stress-strain analysis of rotating discs of nonuniform thickness and density. Tahani *et al.* [13] developed a semi-analytical method for three-dimensional analysis of deformation and stress fields in rotating annular disks made of cylindrically orthotropic nested rings. Afshar [14] performed finite element analysis of the thermoelastic field in a thin circular disk made of FG material subjected to a thermal load and a rotary inertia force. Tutuncu and Temel [15] investigated axisymmetric displacements and stresses in FG hollow cylinders, disks, and spheres subjected to uniform internal pressure, using a layerwise theory and complementary functions method. You *et al.* [16] presented a closed form solution for an annular disk made of materials functionally graded according to a power law function and subjected to a uniform temperature difference. Bayat *et al.* [17] calculated mechanical and thermal stresses in a rotating FG disk with varying thickness subjected to a steady state temperature field. Zenkour [18] derived an exact elasticity solution for FGM annular rotating disks, assuming a plane stress state and exponentially varying material properties. A semi-analytical three-dimensional elasticity analysis for rotating FG disks was carried out by Asghari and Ghafoori [19]. The available two-dimensional plane stress solution was generalized to a three-dimensional one for both hollow and solid disks. Peng and Li [20] presented an analytical method to investigate the steady state thermal stresses in a FG hollow circular disk rotating about its central axis at a constant angular velocity. Zenkour [21]

presented an accurate elastic solution for rotating three layer sandwich solid disks with faces made of different isotropic materials and the core made of FGM.

Dai and Dai [22] solved governing equations of rotating FGM circular disks and presented the distribution of the displacement and stress fields by the Runge-Kutta method. Moreover, some research has been conducted to obtain the thermoelastic stress field in FG rotating disks [23-26]. In addition to the static study, vibration and creep behaviors of FG rotating disks have also been investigated [27-31].

The weight of engineering structures plays a key role in design and manufacture. For this reason, researchers have attempted to study minimization of the weight of rotating disks. In this area, as pioneers, Malkov and Salganskaya [32] used numerical methods for optimization of rotating disks. Cheu [33] studied shape optimization of gas turbine disks with the help of the feasible direction method and sequential linear programming. Using a combination of the boundary element method and sequential linear programming, Zeinkiewics and Campbell [34] presented an optimized version of the curves of engineering structures with the aim of decreasing the stress in the whole body. Jafari and Hojjati [35] minimized the weight of a disk and presented an optimal version of the profile of its geometry using simulated annealing and particle swarm optimization. Adopting the Karush–Kuhn–Tucker method, Jafari [36] presented an optimized response for a disk with variable density through the radial direction. Jafari *et al.* [37] applied three different optimization methods, encompassing Simulated Annealing, *PSO* and Karush-Kuhn-Tucker methods to present an optimized version of weight of rotating disks.

Despite much research in the area of rotating disks, there is still need for effective optimization tools to determine the profile of a disk in order to optimize its weight under certain constraints. In this study, optimization analysis is performed of a FG hollow disk with variable density in the radial direction under the effect of thermo-mechanical loads. An algorithm employing coupled *CPSO* and *DQ* approaches is used to achieve the goal. Evaluation of the algorithm shows that the weight of the FG rotating disk is decreased noticeably more than in previous optimization studies, representing a major advance in this area. In numerical simulation, the effects of gradient index and angular velocity of the FG disk on the distribution of stresses and radial displacement are studied and presented graphically.

2. Problem formulation

2.1. Equations of motion

A hollow disk with the inner radius a and the outer radius b is considered. A schematic view of the geometry is shown in Fig. (1).

The profile of the thickness of the disk varies according to the formula:

$$h = h_0 \left(\frac{r}{b} \right)^{-n} \quad (1)$$

where n is the geometric factor and h_0 denotes the thickness of the disk at the outer radius.

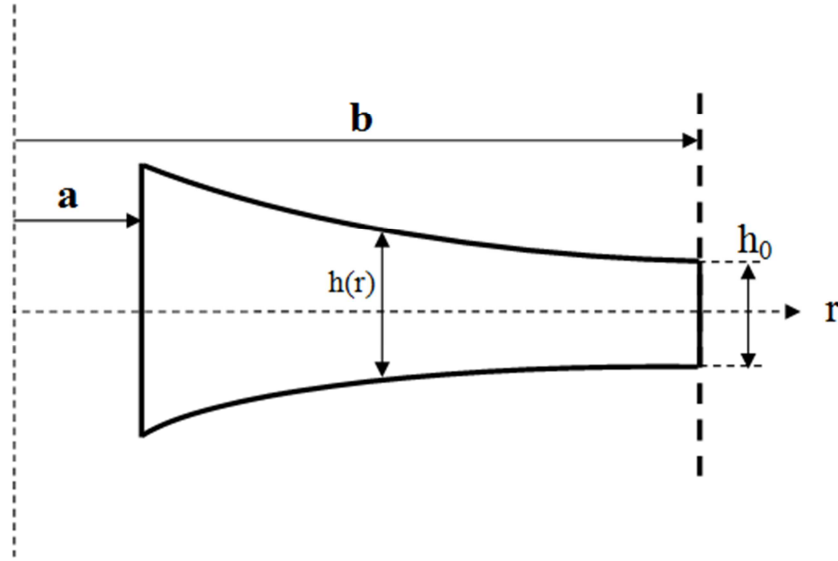


Fig. 1: Schematic view of the cross-section of the disk with varied thickness

The disk is assumed to be rotating at a constant angular velocity (ω) and is subjected to thermal loading, i.e. $T=T(r)$. It is assumed that the disk's material properties of elastic modulus, thermal expansion coefficient, thermal conductivity, and density vary according to a power law through the radial direction as:

$$E = E_0 \left(\frac{r}{b} \right)^p, \quad \alpha = \alpha_0 \left(\frac{r}{b} \right)^p, \quad K = K_0 \left(\frac{r}{b} \right)^p, \quad \rho = \rho_0 \left(\frac{r}{b} \right)^m \quad (2)$$

where E_0 , α_0 , K_0 , and ρ_0 represent the Young's modulus, thermal expansion coefficient, thermal conductivity, and density of the disk at the outer radius, respectively. p and m stand for the gradient indices of properties.

The governing equation of the disk in the presence of a body force can be written as [38]:

$$\frac{d}{dr} (h(r)r\sigma_r) - h(r)\sigma_\theta + h(r)\rho r^2 \omega^2 = 0 \quad (3)$$

Applying the infinitesimal theory of elasticity, the linear strain-displacement relations are:

$$\varepsilon_{rr} = \frac{\partial u}{\partial r} \quad \varepsilon_{\theta\theta} = \frac{u}{r} \quad (4)$$

where u is the displacement component in the radial direction of the rotating disk.

The thermoelastic constitutive equations describing the relation between the strain and stress fields take the form:

$$\begin{aligned} \sigma_r &= \frac{E}{1-\nu^2} \{\varepsilon_r + \nu\varepsilon_\theta - \alpha T(1+\nu)\} \\ \sigma_\theta &= \frac{E}{1-\nu^2} \{\varepsilon_\theta + \nu\varepsilon_r - \alpha T(1+\nu)\} \end{aligned} \quad (5)$$

where ν is Poisson's ratio.

By substituting Eq. (4) into Eq. (5), the components of the stress field are described in terms of displacement and temperature components, and further substitution of these components into Eq. (3) leads to motion equations of the rotating disk with varied thickness in terms of the displacement and temperature parameters of the FG disk:

$$\begin{aligned} &rh(r)E(r)\frac{d^2u}{dr^2} + \{rE(r)\frac{dh(r)}{dr} + rh(r)\frac{dE(r)}{dr} + E(r)h(r)\}\frac{du}{dr} \\ &+ \{vE\frac{dh(r)}{dr} + v h(r)\frac{dE(r)}{dr} - \frac{1}{r}Eh(r)\}u + (1-\nu^2)h(r)\rho(r)r^2\omega^2 \\ &- r(1+\nu)\frac{d(h(r)E(r)\alpha(r)T)}{dr} = 0 \end{aligned} \quad (6)$$

For thermal analysis, steady state heat conduction without an internal heat source is considered [39]:

$$rh(r)K(r)\frac{d^2T}{dr^2} + \{h(r)K(r) + rK(r)\frac{dh(r)}{dr} + rh(r)\frac{dK(r)}{dr}\}\frac{dT(r)}{dr} = 0 \quad (7)$$

2.2. Boundary conditions

It is assumed that the inner surface of the disk is free but the outer radius experiences a uniform thermal shock. Hence, mechanical deflections imposed in the rotating disk are the result of centrifugal force and variation of temperature. Under such circumstances, the corresponding mechanical and thermal boundary conditions are defined as.

$$\begin{aligned} \text{at } r = a: \quad T_a &= 0, \sigma_{r=a} = \frac{E}{1-\nu^2} \{\varepsilon_r + \nu\varepsilon_\theta - \alpha T(1+\nu)\} = 0 \\ \text{at } r = b: \quad T &= T_b, \sigma_{r=b} = \frac{E}{1-\nu^2} \{\varepsilon_r + \nu\varepsilon_\theta - \alpha T(1+\nu)\} = 0 \end{aligned} \quad (8)$$

2.3. Minimizing the weight of the rotating disk

If γ shows specific gravity, we know that the weight of a body can be calculated by the formula:

$$W = \int_{r=a}^{r=b} \gamma dV = \int_{r=a}^{r=b} \rho(r) g dV \quad (9)$$

where $g = 9.81 \text{ m/s}^2$.

Considering Fig. (1), we can derive the volume :

$$dV = 2\pi(2h(r))rdr = 4\pi h(r)rdr \quad (10)$$

By combining Eq. (9) and Eq. (10), the total weight of the rotating disk can be calculated as:

$$W = \int_{r=a}^{r=b} 4\pi h(r) \rho(r) g r dr = 4\pi h_0 \rho_0 g \int_{r=a}^{r=b} \left(\frac{r}{b}\right)^{m-n} r dr \quad (11)$$

Eq. (11) is the goal function in this study, and the values of m and n should be determined to minimize the value of W .

It is assumed that deflections of the rotating disk remain in the elastic region; that being the case, von Mises criteria are applied to guarantee the safety of the disk. Von Mises stress can be determined from:

$$\sigma_e = \sqrt{\sigma_r^2 - \sigma_r \sigma_\theta + \sigma_\theta^2} \quad (12)$$

Now, the optimization problem of the rotating disk can be summarized:

$$\begin{aligned} &\text{Minimize: } \text{Weight} \\ &\text{Constraints: } \sigma_e \leq \sigma_{yield} \quad , \quad n_{min} \leq n \leq n_{max} \quad , \quad p_{min} \leq p \leq p_{max} \end{aligned} \quad (13)$$

In this study, the search domain for n is $[-2, 2]$, (namely, n_{min} and n_{max} are equal to -2 and 2, respectively) and the search space for choosing the gradient index is assumed to vary between $[-1, 1]$.

3. Optimization theory

In most optimization cases in the area of engineering structures, we face a set of goal functions connected with some constraints. A general description of a constrained optimization problem can be derived as:

$$\begin{aligned} &\text{Goal Function: } \text{Minimize } f(x) \\ &\text{Constraints: } g_i(x) \leq 0 \quad , \quad h_i(x) = 0 \\ &\text{Search Space: } x = [x_1, x_2, \dots, x_n] \end{aligned} \quad (14)$$

In the engineering field there are usually two difficulties in optimizing complex problems. The first problem is finding the optimized responses while all constraints are satisfied; the

second is making a connection between the optimization algorithm and the goal function. In this work, to solve the first problem, we use *CPSO* to provide responses under the rule of constraint. Also, the *DQ* method, a powerful approach in the area of solving differential equations, is applied to provide feedback to the goal function.

3.1. Particle swarm optimization

As an evolutionary computation technique, *PSO* has been comprehensively investigated by other researchers [40, 41]. Briefly, some random particles are defined in the search space. Each particle has its own position and velocity, as $x_{i,j}$ and $v_{i,j}$. At every step, each particle obtains its own best position, i.e., $p_{i,j}$, and at every step of the algorithm the global best will be updated ($p_{g,j}$). New positions and velocity of particles can be updated via the formula:

$$\begin{aligned} v_{i,j}(t+1) &= wv_{i,j}(t) + c_1r_1(p_{i,j} - x_{i,j}(t)) + c_2r_2(p_{g,j} - x_{i,j}(t)) \\ x_{i,j}(t+1) &= x_{i,j}(t) + v_{i,j}(t+1) \end{aligned} \quad (15)$$

where c_1 and c_2 are constant; r_1 and r_2 show independent random numbers in $[0, 1]$, and w is the inertia factor. Briefly, Fig. 2 depicts the different steps in the conventional *PSO* method.

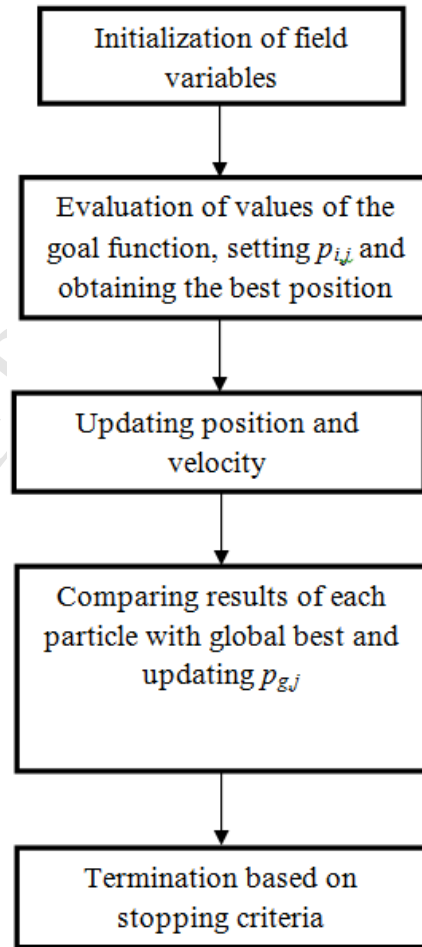


Fig. 2: Flow chart of the *PSO* method

3.2. Co-evolutionary particle swarm optimization

For managing the constraints of the optimization problem, penalty functions are the most popular approach among researchers. For the application of penalty function rules, some improvements have been reported in the literature [42, 43]. In this study, the method proposed in reference [44] is used to meet the constraints. Two simultaneous *PSOs* cooperate interactively; the internal loop searches for variables of optimization in the search space of the problem while the external loop is utilized to insert penalty rules.

The penalty functions for the internal and external *PSO* methods are considered as [45]:

$$\text{Internal } PSO: F_i(x) = f_i(x) + \text{sum_viol} \times w_1 + \text{num_viol} \times w_2$$

$$\text{External } PSO: \begin{cases} \text{-- If there is any feasible solution in internal } PSO: \\ P(B_j) = \frac{\sum f_{\text{feasible}}}{\text{num_feasible}} - \text{num_feasible} \\ \text{-- If there is no feasible solution in internal } PSO: \\ P(B_j) = \max(P_{\text{valid}}) + \frac{\sum \text{sum_viol}}{\sum \text{num_viol}} - \sum \text{num_viol} \end{cases} \quad (16)$$

where $f_i(x)$, sum_viol and num_viol denote objective value, sum of all violations, and number of violations in internal loop, respectively. w_1 and w_2 are penalty factors and are calculated by the external *PSO*. Also, $\sum f_{\text{feasible}}$ and num_feasible show the sum of the objective functions of the feasible solutions and the number of feasible solutions in the internal swarm, respectively, and $\max(P_{\text{valid}})$ is the maximum fitness value of the valid particles in the external swarm. More details of the procedure can be found in [44].

3.3. Differential quadrature method

To solve the coupled equations, including the motion and boundary equations of the elastic and thermal fields, the *DQ* method is utilized to approximate the derivatives of the desired functions in the radial direction. Different versions of the *DQ* method can be found in the literature; in this work, we apply one of the most useful approaches that uses the following Lagrange interpolation polynomials as the test function:

$$g_k(x) = \frac{M(x)}{(x - x_k)M^{(1)}(x_k)}, k = 1, 2, \dots, N \quad (17)$$

where:

$$M(x) = (x - x_1)(x - x_2) \dots (x - x_N) \quad , \quad M^{(1)}(x_k) = \prod_{\substack{k=1 \\ k \neq i}}^N (x - x_k) \quad (18)$$

By applying the above equations at N grid points, algebraic formulations to compute the weighting coefficients of the first, and second order derivatives of the function $f(x)$, are obtained [46]:

$$A^{(1)}_{ij} = \frac{1}{x_j - x_i} \prod_{\substack{k=1 \\ k \neq i}}^N \frac{x_i - x_k}{x_j - x_k} \quad \text{when } i \neq j, \quad A^{(1)}_{ii} = \sum_{\substack{k=1 \\ k \neq i}}^N \frac{1}{x_i - x_k}, \quad (19)$$

$$A^{(2)}_{ij} = \sum_{k=1}^N A^{(1)}_{il} A^{(1)}_{il}$$

where $A^{(1)}$ and $A^{(2)}$ denote the weighting coefficients of the first and the second order derivatives respectively of the function $f(x)$. Hence, the first order and the second order derivatives of the desired function in the polar coordinates are approximated as:

$$\left. \frac{\partial f}{\partial r} \right|_i \approx \sum_{l=1}^N A^{(1)}_{il} f_l, \quad \left. \frac{\partial^2 f}{\partial r^2} \right|_i \approx \sum_{l=1}^N A^{(2)}_{il} f_l \quad (20)$$

By applying the above formulation, the governing equation of motion is then discretized into DQ forms as:

$$\begin{aligned} & r_i h(r) E(r) \sum_{l=1}^N A^{(2)}_{il} u_l + \{r_i E(r) \frac{dh(r)}{dr} + r_i h(r) \frac{dE(r)}{dr} + E(r) h(r)\} \sum_{l=1}^N A^{(1)}_{il} u_l \\ & + \{v E(r) \frac{dh(r)}{dr} + v h(r) \frac{dE(r)}{dr} - \frac{1}{r_i} E(r) h(r)\} u_i + (1 - v^2) h(r) \rho(r) r_i^2 \omega^2 \\ & - r_i (1 + v) \left\{ \frac{d\{h(r) E(r) \alpha(r)\}}{dr} T_i + h(r) E(r) \alpha(r) \sum_{l=1}^N A^{(1)}_{il} T_l \right\} = 0 \end{aligned} \quad (21)$$

Derivatives of the elastic constant E , the thermal expansion coefficient α , and the thickness of the FG disk are defined with reference to their respective functions. Variation of the thermal field occurs throughout the radial direction of the disk; hence, the steady state temperature field is defined as:

$$r_i h(r) K(r) \sum_{l=1}^N A^{(2)}_{il} T_l + \{h(r) K(r) + r_i K(r) \frac{dh(r)}{dr} + r_i h(r) \frac{dK(r)}{dr}\} \sum_{l=1}^N A^{(1)}_{il} T_l = 0 \quad (22)$$

In Eq. (22), the first order derivative of the thermal conductivity, K , is determined according to Eq. (2). Furthermore, the stress and displacement boundary conditions are defined as:

$$\sigma_{r=a,b} = \frac{E}{1-\nu^2} \left\{ \sum_{l=1}^N A_{il}^{(1)} u_l + \nu \frac{u_i}{r_i} - (1+\nu) \alpha(r) T(r) \right\} = 0 \quad (23)$$

4. Results and discussion

The results are presented in dimensionless form by defining the following dimensionless stress and displacement fields:

$$\begin{Bmatrix} \bar{\sigma}_r \\ \bar{\sigma}_\theta \\ \bar{\sigma}_\nu \end{Bmatrix} = \frac{1}{\sigma_{yield}} \begin{Bmatrix} \sigma_r \\ \sigma_\theta \\ \sigma_\nu \end{Bmatrix}, \quad \bar{u} = \frac{1}{b} u \quad (24)$$

4.1. Evaluation of the present method

To evaluate the accuracy of the present method, a study is performed to compare its results with optimized results reported in literature [37]. It is assumed that the disk rotates at $\omega=500$ (rad/s), the inner and outer radii of the disk are $a=0.1$ (m) and $b=0.6$ (m), respectively; and the thickness of the disk at $r=b$ is equal to 0.1 (i.e., $h_0=0.1$). In this case, the elastic modulus and Poisson's ratio are constant throughout the thickness, but the density of the disk varies according to Eq. (2). The material properties of the rotating disk are given in Table 1.

Table 1: Material properties of stainless steel at the outer surface of the disk [37]

Material properties	Value
E (GPa)	207
ν	0.3
α ($10^{-6}/^\circ\text{C}$)	1.2×10^{-6}
K (W/m. $^\circ\text{C}$)	20
ρ (kg/m^3)	7850
σ_{yield} (MPa)	300

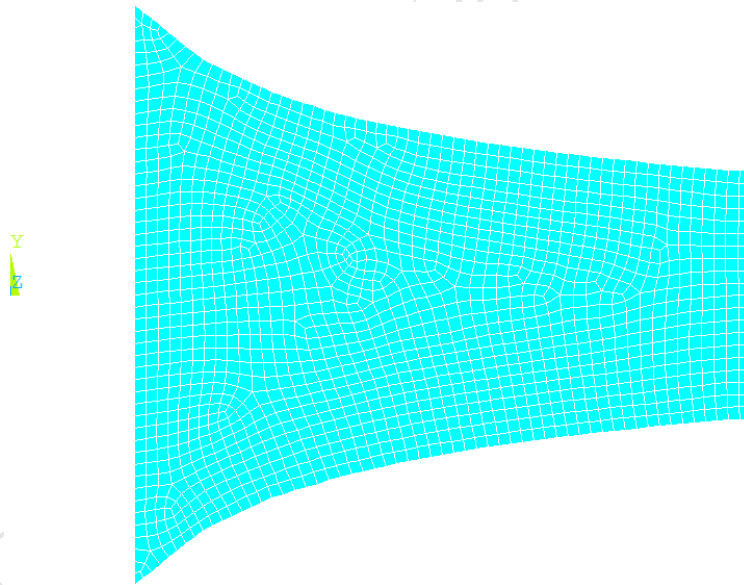
The weight reduction results of the disk are given in Table 2.

Table 2: Comparison between present results and reference [37]

		Weight of Disk (N)	Weight Reduction
Initial Weight		29016.802	-
Reference [37]	SA method	15982.4	45%
Reference [37]	PSO method	14658.2	50%
Reference [37]	KKT method	14168.2	51%
Present study		13392.143	53.8%

As can be seen, when compared with Ref. [37], the results from the coupled *CPSO-DQ* show a much better percentage of weight reduction, presenting a dramatic improvement while guaranteeing the disk's safety ($\sigma_v \leq \sigma_{yield}$).

Furthermore, a comparison between results of this study with those of the finite element analysis is shown in Fig. 3. Those results are in good agreement, demonstrating accuracy of the presented approach.



(a)

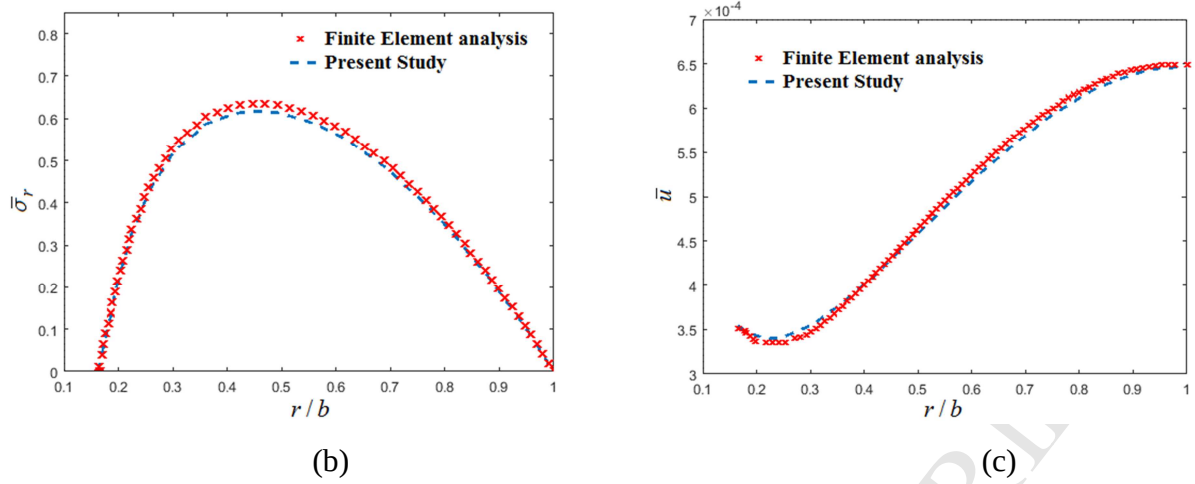


Fig. 3: Verifying results with those of the finite element analysis (a) finite element mesh (b) radial stress and (c) radial displacement

4.2. Investigation of the effects of the gradient index

It is assumed that variations of the modulus of elasticity, the coefficient of thermal expansion, and the thermal conduction coefficient are graded in the radial direction according to Eq. (2). In many engineering materials, the Poisson's ratio varies slightly; hence it is assumed to be a constant throughout the disk. The rotating hollow disk is under the effect of centrifugal force as well as thermal loading. Temperatures at the inner and the outer radii are assumed to be equal to $T(a)=0$ °C and $T(b)=100$ °C, respectively. Considering the constraint $\sigma_v \leq \sigma_{yield}$, the FG annulus disk rotates at a constant angular velocity of $\omega=500$ rad/s and the numerical results for different values of the gradient index, i.e. p , are given in Table 3. As is evident, the higher the value of the gradient index, the higher the percentage of weight reduction.

Table 3: Results of the optimization for different gradient indices

	Disk Profile	Disk Weight (N)	Weight Reduction
Initial weight	$n=1$	29031.52	-
$p=-0.2$	$n=0.6347$	14516.09	50.0%
$p=-0.1$	$n=0.5610$	14102.96	51.4%
$p=0$	$n=0.4775$	13660.40	52.9%
$p=0.1$	$n=0.3853$	13200.80	54.5%
$p=0.2$	$n=0.2920$	12764.28	56.0%

For different values of the gradient index, optimized profiles of the rotating disk throughout the radial direction are shown in Fig. 4.

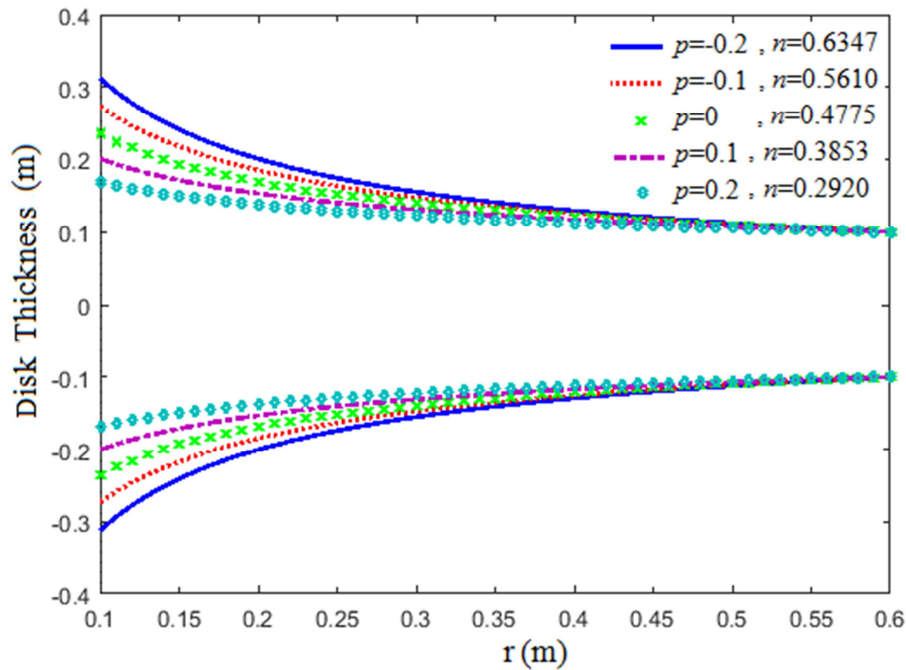


Fig. 4: Optimized profiles of the rotating FG disk for different values of the gradient index

Fig. 5 represents the effects of the gradient index on the stress and displacement fields. From Fig. 5a it is evident that change in the gradient index has neglectable impact on the radial component of the stress field. Also, the dimensionless radial stress reaches zero value at the inner and the outer surfaces, showing that the boundary conditions are satisfied perfectly. Figs. 5b and 5c demonstrate that, as the gradient index increases, the circumferential and von Mises stresses increase rapidly. The dimensionless circumferential and von Mises stresses are reduced by moving from the inner boundary towards the outer boundary; hence, the maximum values of these stresses occur at the inner surface of the disk. It is also interesting to note that the rate of change in the dimensionless circumferential and von Mises stresses at the middle surface of the rotating disk is lower than that of sections near the inner and outer boundaries. The influence of the grading parameter on dimensionless radial displacement is shown in Fig. 5d, indicating a monotonous increase with the increment of p . The maximum value of dimensionless displacement occurs at the outer surface. These results provide useful information for the optimal design of FG annulus disks for detecting probable fracture regions in the structure.

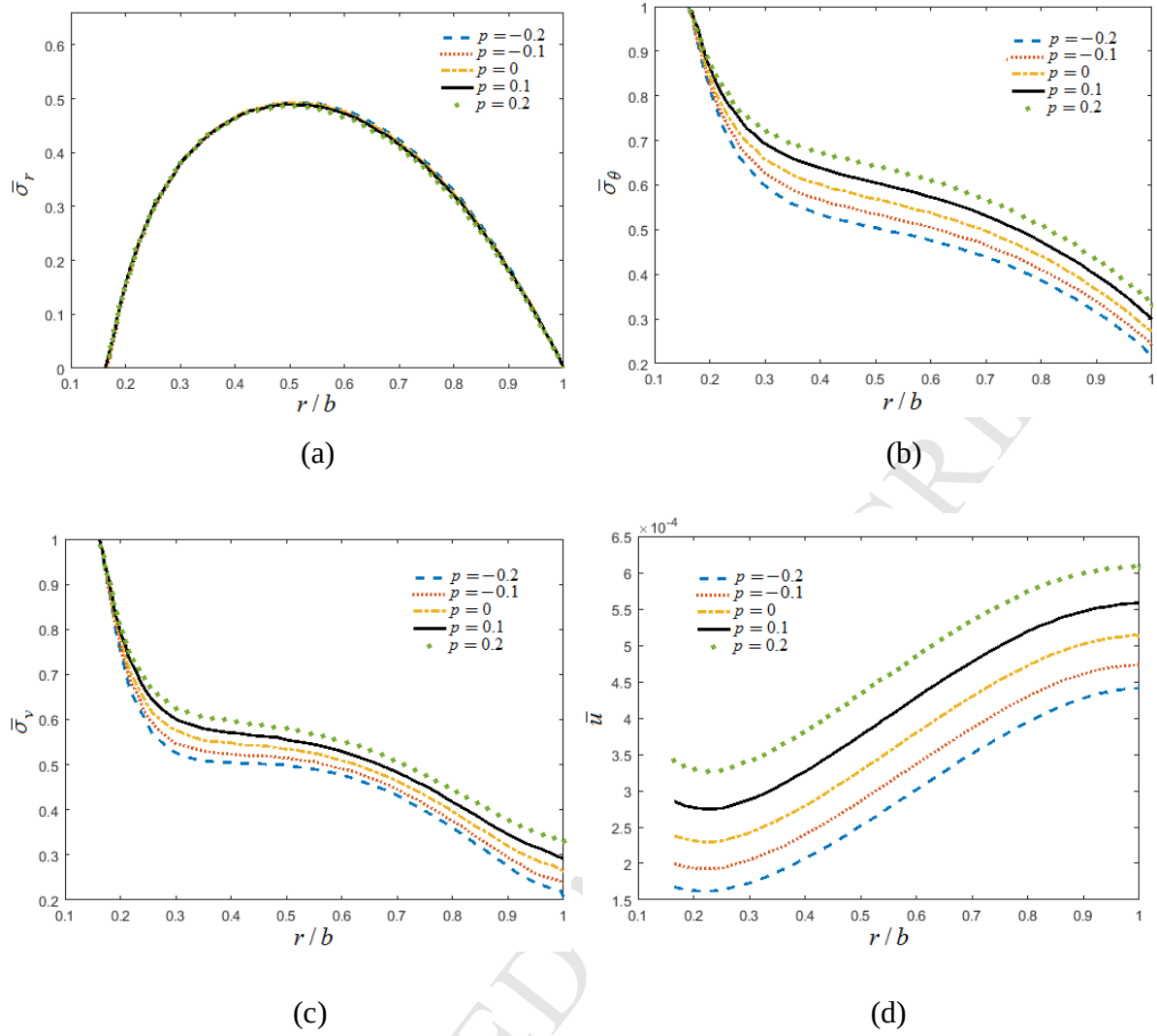


Fig. 5: Effects of gradient index on the radial distribution of (a) radial stress, (b) circumferential stress, (c) Von Mises stress, (d) radial displacement

4.3. Investigation of impacts of angular velocity

Finally, the effects of variation of the angular velocity ω on the distribution of optimized responses of stress and displacement fields of a FG disk with $p=0.1$ are investigated. The FG rotating disk considered is subjected to thermal loadings of $T(a)=0$ °C and $T(b)=100$ °C at the inner and the outer boundaries respectively. For different values of angular velocity, the optimized profiles of the rotating FG disk are obtained, as shown in Fig. 6.

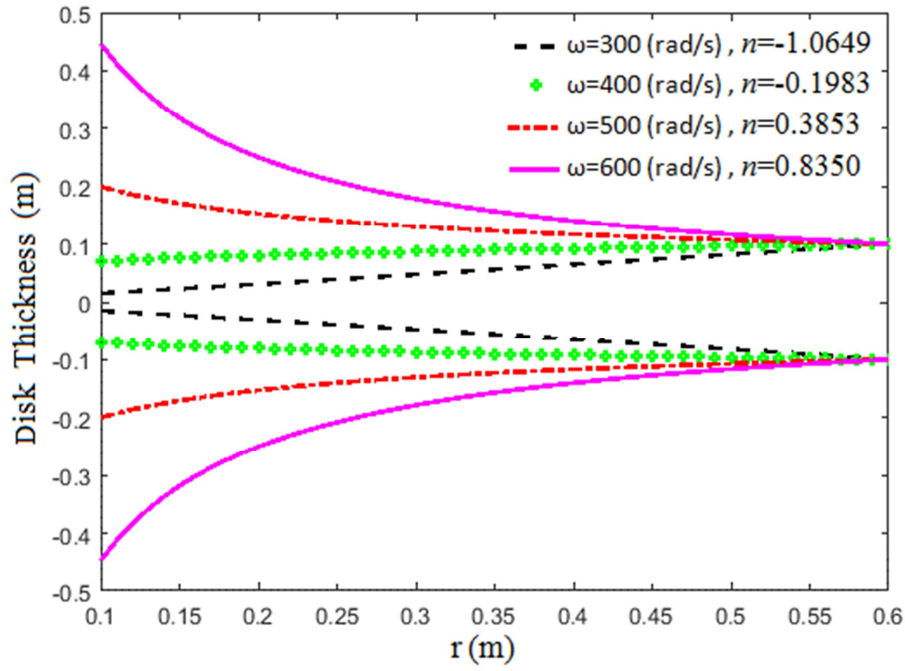


Fig. 6: Optimized profiles of rotating FG disk for different values of angular velocity

Table 4: Results of optimization for different angular velocities

	Disk Profile	Disk Weight (N)	Weight Reduction
Initial weight	$n=1$	29031.52	-
$\omega=300$ (rad/s)	$n=-1.0649$	8564.51	70.5%
$\omega=400$ (rad/s)	$n=-0.1983$	10857.26	62.6%
$\omega=500$ (rad/s)	$n=0.3853$	13200.80	54.5%
$\omega=600$ (rad/s)	$n=0.8350$	15758.79	45.7%

As ω changes, the thermoelastic responses of the stress and displacement fields are shown in Fig. 7. As shown in Fig. 7a, the dimensionless radial stress increases with the increase of ω . It is noteworthy that for a small angular velocity (e.g., $\omega=300$), the maximum value of the dimensionless radial stress occurs at a position closer to the inner boundary, whereas for a large value of ω , e.g., $\omega=600$, the position shifts away from the inner boundary to a position closer to the middle of the disk. The radial distribution of the circumferential and von Mises stresses are plotted in Figs. 7b and 7c respectively, indicating that their distribution profiles are highly sensitive to the rotating angular velocity, with stresses increasing significantly as ω increases. The variation of the dimensionless radial displacement with angular velocity is

plotted in Fig. 7d. As expected, the dimensionless radial displacement increases monotonously as ω increases.

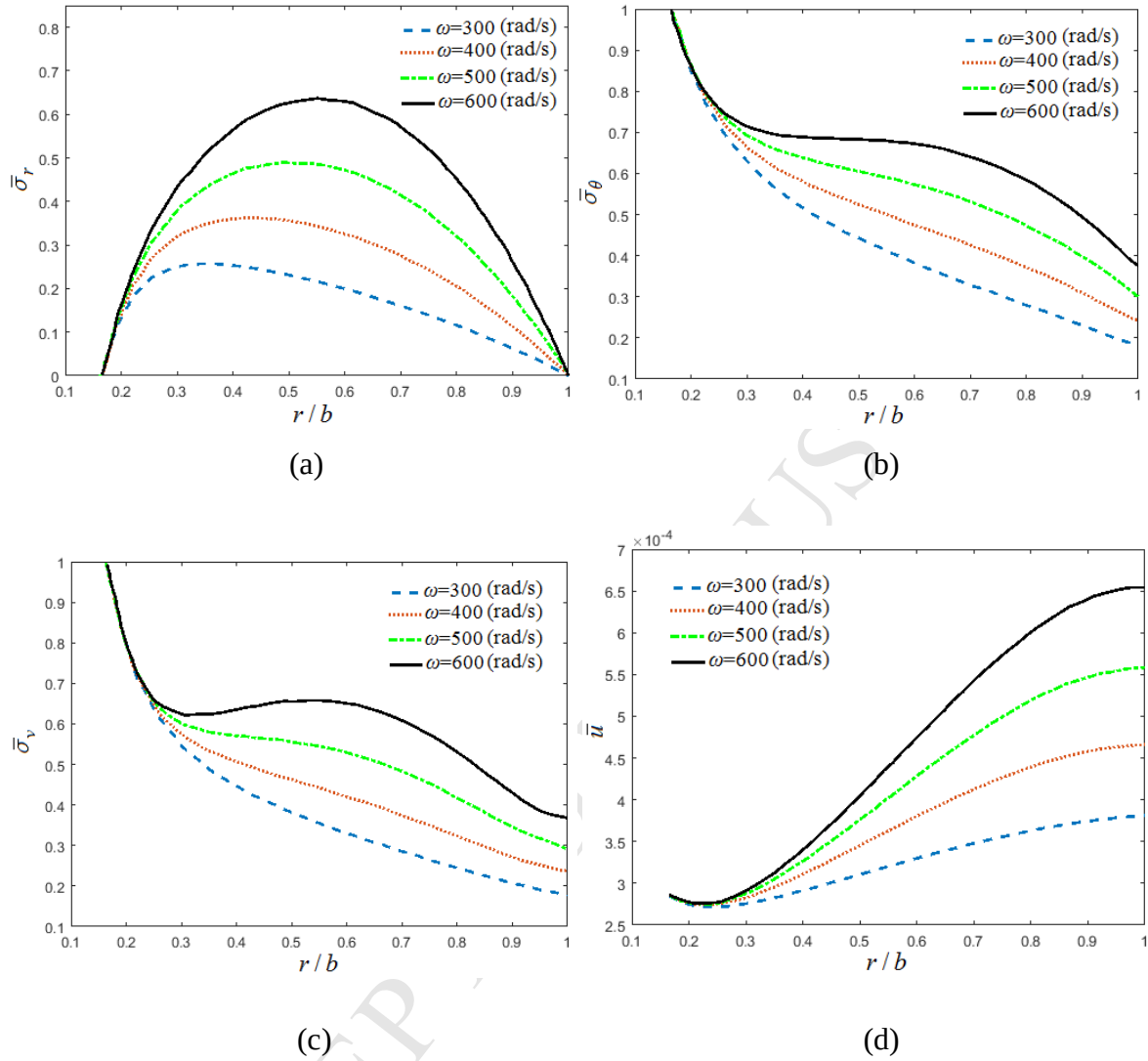


Fig. 7: Effects of angular velocity on the radial distribution of (a) radial stress, (b) circumferential stress, (c) Von Mises stress, (d) radial displacement

5. Conclusion

In this study, a combination of *CPSO* and the *DQ* approach was used to optimize the thermoelastic response of FG disks of varied thickness. The material properties of the disk, namely elastic modulus, thermal expansion coefficient, thermal conductivity, and density, changed through the radial direction according to a power law. The *CPSO* method was applied to search in the search space for design variables and the *DQ*, as a powerful

numerical method, worked to discretize and solve the thermoelastic governing equations of the disk. The effectiveness of the present approach was confirmed by comparing the numerical results with previous optimized work in the literature, with considerable improvement observed. The optimized profiles for different values of gradient index and angular velocity were obtained, and the distribution of dimensionless stress and displacement fields of the rotating hollow disk were calculated and presented. The circumferential and equivalent von Mises stresses are practically more important, because cracking and yielding of circular disks generally occurs when the dimensionless circumferential and von Mises stress reach the determined critical stress levels for brittle and ductile materials, respectively. Hence, the findings presented in this study can help designers choose an optimal version of a FG rotating disk according to the appropriate grading parameter.

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References

- [1] U. Gramer, The elastic-plastic stress distribution in the rotating annulus and in the annulus under external pressure, *J. Appl. Math. Mech.* 64 (1984) 126-128.
- [2] U. Gramer, Distribution in the rotating elastic-plastic disk, *J. Appl. Math. Mech.* 65 (1985) 136-137.
- [3] U. Güven, The fully plastic rotating disk of variable thickness, *J. Appl. Math. Mech.* 74 (1994) 61-65.
- [4] U. Güven, On the applicability of Tresca's yield condition to the linear hardening rotating solid disk of variable thickness, *J. Appl. Math. Mech.* 75 (1995) 397-398.
- [5] M. Koizumi, Functionally gradient materials the concept of FGM, *Ceramic Transactions* 34 (1993) 3-10.
- [6] S. Suresh, A. Mortensen, *Fundamentals of functionally graded materials*, Institute of Materials (IOM) Communications Limited, London, 1998.
- [7] J.N. Reddy, C.M. Wang, S. Kitipornchai, Axisymmetric bending of functionally graded circular and annular plate, *Eur. J. Mech. A-SOLID* 18 (1999) 185-199.
- [8] Y. Fukui, N. Yamanaka, Elastic analysis for thick-walled tubes of functionally graded material subjected to internal pressure, *JSME Int. J. Ser. 1, Solid Mech., Streng. Mat.* 35(1992) 379-385.

- [9] M.R. Eslami, M.H. Babaei, R. Poultangari, Thermal and mechanical stresses in a functionally graded thick sphere, *Int. J. Pres. Ves. Pip.* 82 (2005) 522-527.
- [10] M.H. Hojjati, S. Jafari, Variational iteration solution of elastic non uniform thickness and density rotating disks, *Far East J. Appl. Math.* 29 (2007) 185-200.
- [11] M.H. Hojjati, S. Jafari, Semi exact solution of elastic non uniform thickness and density rotating disks by homotopy perturbation and adomian's decomposition methods Part I: elastic solution, *Int. J. Pres. Ves. Pip.* 85 (2008) 871-878.
- [12] M.H. Hojjati, A. Hassani, Theoretical and numerical analyses of rotating discs of nonuniform thickness and density, *Int. J. Pres. Ves. Pip.* 85 (2008) 694-700.
- [13] M. Tahani, A. Nosier, S.M. Zebarjad, Deformation and stress analysis of circumferentially fiber-reinforced composite disks, *Int. J. Solids Struct.* 42 (2005) 2741-2754.
- [14] A.M. Afsar, Finite element analysis of thermoelastic field in a rotating FGM circular disk, *Appl. Math. Model.* 34 (2010) 3309-3320.
- [15] N. Tutuncu, B. Temel, A novel approach to stress analysis of pressurized FGM cylinders, disks and spheres, *Comput. Struct.* 91 (2009) 385-390.
- [16] L.H. You, X.Y. You, J.J. Zhang, J. Li, On rotating circular disks with varying material properties, *J. Appl. Math. Phys.* 58 (2007) 1068-1084.
- [17] M. Bayat, M. Saleem, B.B. Sahari, A.M.S. Hamouda, E. Mahdi, Mechanical and thermal stresses in a functionally graded rotating disk with variable thickness due to radially symmetry loads, *Int. J. Pres. Ves. Pip.* 86 (2009) 357-372.
- [18] A.M. Zenkour, Analytical solutions for rotating exponentially-graded annular disks with various boundary conditions, *Int. J. Struct. Stab. Dy.* 5 (2005) 557-577.
- [19] M. Asghari, E. Ghafoori, A three-dimensional elasticity solution for functionally graded rotating disks, *Comput. Struct.* 92 (2010) 1092-1099.
- [20] X.L. Peng, X.F. Li, Thermal stress in rotating functionally graded hollow circular disks, *Comput. Struct.* 92 (2010) 1896-1904.
- [21] A.M. Zenkour, Stress distribution in rotating composite structures of functionally graded solid disks, *J. Mater. Process. Technol.* 209 (2009) 3511-3517.
- [22] T. Dai, H.-L. Dai, Investigation of mechanical behavior for a rotating FGM circular disk with a variable angular speed, *J Mech Sci Technol*, 29 (2015) 3779-3787.
- [23] M. Damircheli, M. Azadi, Temperature and thickness effects on thermal and mechanical stresses of rotating FG-disks, *J. Mech. Sci. Technol.* 25 (2011) 827-836.

- [24] A. Hassani, M.H. Hojjati, E. Mahdavi, R. A. Alashti, G. Farrahi, Thermo-mechanical analysis of rotating disks with non-uniform thickness and material properties, *Int. J. Press. Vessels. Pip.* 98 (2012) 95-101.
- [25] A.M. Afsar, J. Go, Finite element analysis of thermoelastic field in a rotating FGM circular disk, *Appl. Math. Model.* 34 (2010) 3309-3320.
- [26] A.M. Afsar, J. Go, J.I. Song, A mathematical analysis of thermoelastic characteristics of a rotating circular disk with an FGM coating at the outer surface, *Adv. Compos. Mater.* 19 (2010) 269-288.
- [27] J.N. Sharma, D. Sharma, S. Kumar, Vibration analysis of a rotating FGM thermoelastic axisymmetric circular disk using FEM, *Int. J. Comput. Methods. Eng. Sci. Mech.* 14 (2013) 262-270.
- [28] H. Bahaloo, J. Papadopolus, R. Ghosh, E. Mahdi, A. Vaziri, HamidNayeb-Hashemia Transverse vibration and stability of a functionally graded rotating annular disk with a circumferential crack, *Int. J. Mech. Sci.* 113 (2016) 26-35.
- [29] S.A. Hosseini Kordkheili, M. Livani, Thermoelastic creep analysis of a functionally graded various thickness rotating disk with temperature-dependent material properties, *Int. J. Press. Vessels. Pip.* 111-112 (2013) 63-74.
- [30] M. Rattan, N. Chamoli, S.B. Singh, N. Gupta, Creep behavior of anisotropic functionally graded rotating discs, *Int. J. Comput. Mater. Sci. Eng.* 02 (2013) 1350005.
- [31] K. Khanna, V.K. Gupta, S.P. Nigam, Creep analysis of a variable thickness rotating FGM disc using tresca criterion, *Def. Sci. J.* 65 (2015) 163-170.
- [32] V.P. Malkov, E.A. Salgankays, Optimum material distribution in rotating disks for minimum strength, *Sov Aeronaut.* 19 (1976) 46-50.
- [33] T. Cheu, Procedures for shape optimization of gas turbine, *Comput. Struct.* 54 (1990) 1-4.
- [34] O.C. Zeinkiewics, J.S. Campbell, Shape optimization and sequential linear programming in optimum structural design, Wiley, New York, 1973.
- [35] S. Jafari, M.H. Hojjati, Modern optimization methods in minimum weight design of elastic annular rotating disk with variable thickness, *AIP Conference Proceedings* 1400, 359 (2011); doi: 10.1063/1.3663143.
- [36] S. Jafari, The Karush-Kuhn-Tucker optimality conditions in minimum weight design of elastic rotating disks with variable thickness and density, *Int. J. Indust. Eng. Comput.* 2 (2011) 765-774.

- [37] S. Jafari, M.H. Hojjati, A. Fathi, Classical and modern optimization methods in minimum weight design of elastic rotating disk with variable thickness and density, *Int. J. Pres. Ves. Pip.* 92 (2012) 41-47.
- [38] S.P. Timoshenko, J.N. Goodier, *Theory of elasticity*, 3rd ed., McGrawHill, Singapore, 1984.
- [39] M. Bayat, B.B. Sahari, M. Saleem, Aidy Ali, S.V. Wong, Thermoelastic solution of a functionally graded variable thickness rotating disk with bending based on the first-order shear deformation theory, *Thin-Walled Struct.* 47 (2009) 568-582.
- [40] J. Kennedy, R.C. Eberhart, Particle swarm optimization. In: *Proceedings of the 1995 IEEE International Conference on Neural Networks*, IEEE Service Center, Piscataway, NJ, (1995) 1942-1948.
- [41] J. Kennedy, R.C. Eberhart, R.C., Y. Shi, *Swarm Intelligence*, Morgan Kaufmann, San Francisco, 2001.
- [42] Coello, C.A.C., 2000. Use of a self-adaptive penalty approach for engineering optimization problems, *Comput. Indust.* 41 (2001) 113-127.
- [43] Z. Michalewicz, N. Attia, Evolutionary optimization of constrained problems. In: Sebald, A.V., Fogel, L.J. (Eds.), *Proceedings of the third Annual Conference on Evolutionary Programming*. World Scientific, River Edge (1994) 98-108.
- [44] Q. He, L. Wang, An effective co-evolutionary particle swarm optimization for constrained engineering design problems, *Eng. Appl. Artif. Intel.* 20 (2007) 89-99.
- [45] J.T. Richardson, M.R. Palmer, G. Liepins, M. Hilliard, Some guidelines for genetic algorithms with penalty functions. In: Schaffer, J.D. (Ed.), *Proceedings of the Third International Conference on Genetic Algorithms*, George Mason University, Morgan Kaufmann, (1989) 191-197.
- [46] C. Shu, *Differential quadrature and its application in engineering*, Springer-Verlag, London, 2000.

Optimization of weight of functionally graded rotating disks of variable thickness under thermoelastic load

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Highlights:

- An effective design algorithm for functionally graded rotating disks
- Weight optimization of functionally graded rotating disks with different thickness
- Thermal effects are considered as well as those of the centrifugal impacts
- Optimized profiles of functionally graded disks with different angular velocities and grading indexes